

Case Study 1.7: When the loop dies: Getting on with MDI

V 1.3



This case study reports a user experience in a 20 day period of having no access to a closed loop (due to a broken AAPS compatible pump).

Data were collected in Nightscout after a Combo E7 error (pump break-down) at the end of a Spain vacation

Method

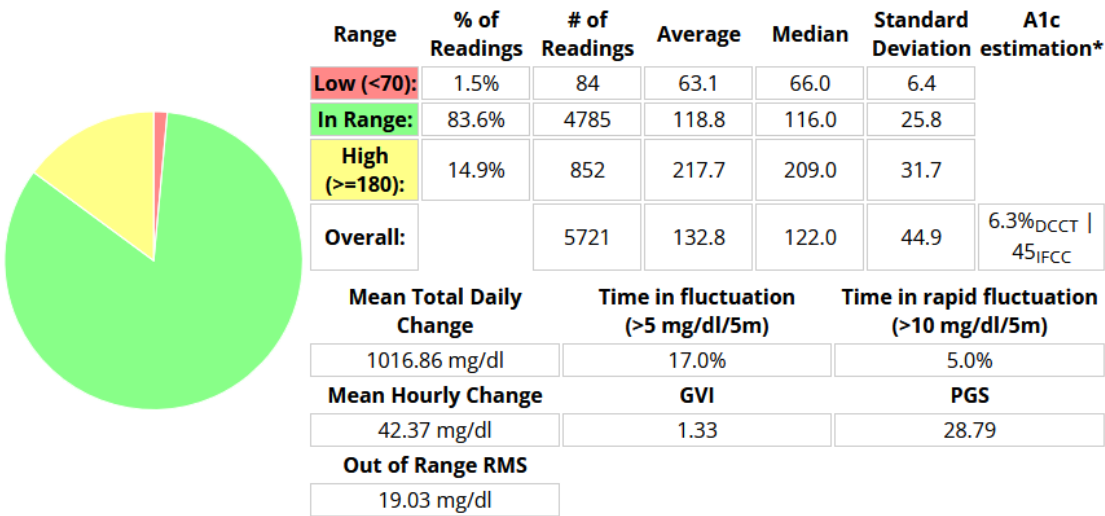
Adult on Lyumjev , TDD ~ 37 U of which 17 U profile basal; ISF ~ 30 mg/dl/U; IC ~ 8 g/ U

- A. Full Closed Loop (FCL = no bolusing and no carb inputs by the user):
2x G6; Combo pump; AAPS/autoISF 3.2.0.4/3.0.1
- B. Bolusing for meals and for corrections;
15 of 20 days using an old pump to give profile basal. Virtual Pump mode in AAPS (Open Loop, entering insulin boli, and estimating carbs; “ MDI+ ”)

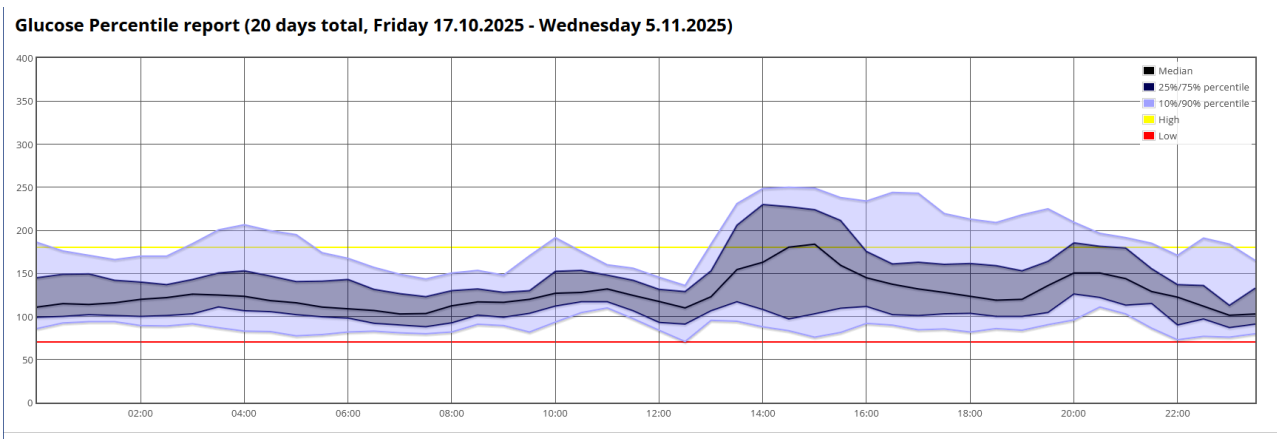
Results with Method B: Bolusing with pen + AAPS in Virtual Pump setting (MDI+)

20 days (17 Oct – 05 Nov. 2025) , mostly at home, without closed loop (after Combo E7 break-down).

Glucose distribution (20 days total, Friday 17.10.2025 - Wednesday 5.11.2025)



24 **TIR percentile chart** 20 days wiithout closed loop:



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27 **MIN and MAX daily glucose values**

28 20 days wiithout closed loop:

Date	Min	Max
Wednesday 5.11.2025	64	204
Tuesday 4.11.2025	62	235
Monday 3.11.2025	49	301
Sunday 2.11.2025	56	185
Saturday 1.11.2025	56	166
Friday 31.10.2025	76	212
Thursday 30.10.2025	57	226
Wednesday 29.10.2025	81	169
Tuesday 28.10.2025	71	242
Monday 27.10.2025	66	320
Sunday 26.10.2025	72	269
Saturday 25.10.2025	68	237
Friday 24.10.2025	39	268
Thursday 23.10.2025	63	277
Wednesday 22.10.2025	77	188
Tuesday 21.10.2025	67	176
Monday 20.10.2025	50	229
Sunday 19.10.2025	55	173
Saturday 18.10.2025	70	275
Friday 17.10.2025	59	291
< 60 or > 250	8 / 20	7 / 20

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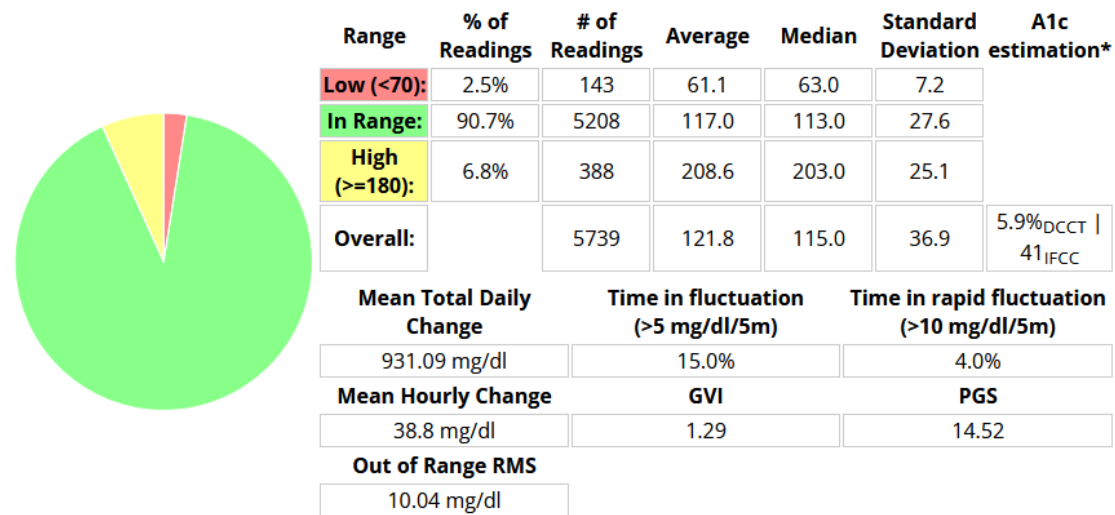
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Results with Method A (AAPS Full Closed Loop, FCL)

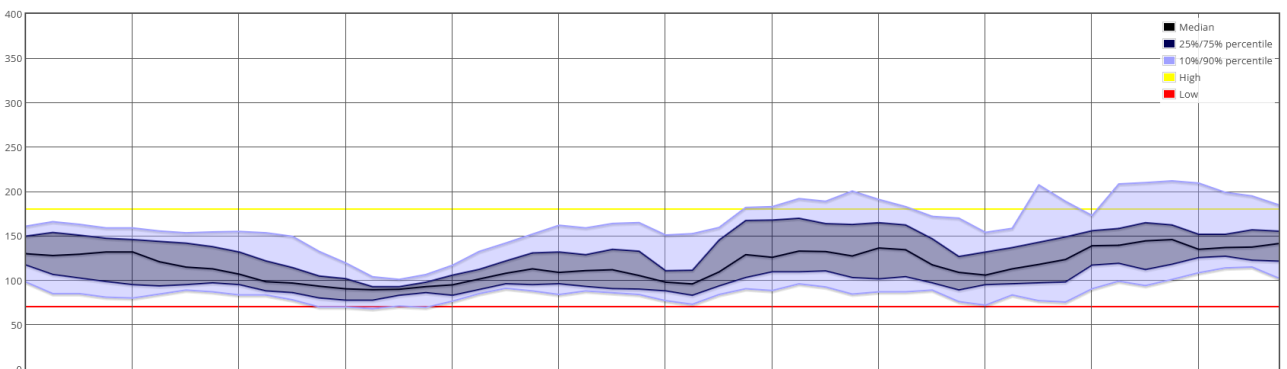
20 prior days in Full Closed Loop (half of which in Spain vacation)

Glucose distribution (20 days total, Saturday 27.9.2025 - Thursday 16.10.2025)



The Full Closed Loop period included 8 vacation days => a bit more challenging for the FCL

Glucose Percentile report (20 days total, Saturday 27.9.2025 - Thursday 16.10.2025)



MIN and MAX daily glucose values

Thursday 16.10.2025	60	235
Wednesday 15.10.2025	69	221
Tuesday 14.10.2025	59	271
Monday 13.10.2025	72	196
Sunday 12.10.2025	70	211
Saturday 11.10.2025	75	227
Friday 10.10.2025	54	214
Thursday 9.10.2025	70	169
Wednesday 8.10.2025	70	154
Tuesday 7.10.2025	56	202

Monday 6.10.2025	90	230
Sunday 5.10.2025	56	221
Saturday 4.10.2025	70	219
Friday 3.10.2025	52	240
Thursday 2.10.2025	41	280
Wednesday 1.10.2025	39	184
Tuesday 30.9.2025	48	181
Monday 29.9.2025	56	177
Sunday 28.9.2025	51	185
Saturday 27.9.2025	56	185

39 < 60 or > 250 **11** / 20 **1** / 20

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41 Discussion

42 This is a comparison of 20 days on a multiple daily injections (MDI+) strategy that was
43 mostly guided by AAPS data in Virtual Pump / Open Loop mode, with 20 preceding days on
44 autoISF Full Closed Loop (FCL).-

45 **High bg values** (post-prandial bg rises; roller-coasters)

46 MDI could actually be superior to FCL because of the following inherent principal advantages
47 vs FCL:

48 1) Pre-bolusing for meals should limit post-prandial bg rises

49 2) Counting and entering carbs, plus inputting bolus information into AAPS, plus watching
50 insulin activity curve vs est. carb digestion, should allow an experienced user to navigate
51 quite well.

52 Indeed, 83.6 % TIR (HbA1c 6.3%) is probably an acceptable situation.

53 It is significantly worse, though, than what resulted in the 20 preceding days using FCL,
54 showing 90.7 % TIR (HbA1c 5.9%). *)

55 *) This FCL period included vacation, 24/7 with friends, different meal and activity patterns than usual.
56 The user's long term average FCL TIR is 93-94%, with lower lab confirmed HbA1c, too.

57 So, what are the disadvantages of the MDI strategy vs. FCL leading to much higher number
58 of values over 250 mg/dl?:

59 1) In daily life, navigating the bg cannot be done several times each hour (while the FCL can
60 do it every 5 minutes).

61 2) Specifically, it is very difficult to manage meals with more than one bolus given, as usually
62 required in MDI (and especially so when using a "fast" insulin.- Rough estimates (size,

timing) for further bolusing after the initial meal bolus very often make the bg run low or high. This can be improved when using data visible in AAPS Open Loop / Virtual Pump mode. But nobody can consistently payi close attention for hours after each meal. That is an advantage that only the automatic management in Closed Loop achieves easily.

3) In MDI there is no night regulation, except alarms alerting for extremes. This results in a wide spread of values in the night data. In contrast, it is one of the easy achieved benefits of closed looping that the loop usually does an excellent job to provide bg in a narrow good range at the beginning of each day. See spread between 4 and 8 am in both Glucose Percentile reports).

Low bg values (under 60 mg/dl)

In FCL, there were 11 of 20 days with bg values under 60 mg/dl, while in MDI there was an incidence of 8 / 20 days.

This is not a worrisome result in FCL, where swinging low is always counter-regulated, meaning usually no user intervention is needed because it self-resolves.

The 8/20 days in MDI is more of a concern. It actually included events where a bg roller coaster had developed, and emergency fast carbs were required (and on going activities had to be interrupted) to prevent a dangerous hypo.

Conclusion

MDI+ performed inferior to FCL.

More remarks on (not) employed methods

"MDI+" vs MDI

Many years back the author had managed his T1D with classical MDI management, first with 2 types of insulin and pens plus a glucometer, later replacing the retard insulin with a pump to provide for the basal needs.

The advent of CGM systems was really not a decisive plus for quality of MDI treatment because it "invited" rash decisions, with ensuing roller-coasters.

Aside from trying to micromanage daily bg in MDI, CGMs did and do of course provide great benefits. For instance, just by periodically analyzing bg developments with the aid of a CGM will support behavior changes (eating, moving around...) that help improve bg curves . Likewise, CGMs are invaluable for

determining the basal rate, especially the nightly one. And all Closed Loops absolutely rely on a good CGM 24/7.

Also the more rapid insulins that came to market did not really make a big positive difference for MDI. OK, they can help in case you forgot to bolus, or if you can't pre-bolus because, in a restaurant, you have no control over when the food will actually arrive. But, if you properly pre-bolus, you might actually benefit from using a **slow** insulin because its activity matches better with carb absorption (digestion), and there is reduced need for second and third boli, which are typically hard to figure out in terms of both, size and timing. Incidence and severity of roller-coasters therefore might be reduced when using a "slower" insulin.

For a really improved MDI therapy (" **MDI+** "), the user must understand the sluggish nature of control. This is greatly, and in real-time, supported by employing AAPS in Virtual Pump/Open Loop mode (...which does require a good CGM).

If the user inputs all the given insulin manually, the AAPS home screen displays the aggregate insulin activity from past bolusing, and even looks into the next hour or two. Carb decay can be realistically modelled, as well. Over time, the user learns, when and how to interfere for desired results – just by watching the curves develop, and also from "Open Loop" suggestions made by AAPS.

Would Hybrid Closed Loop (HCL) offer even better %TIR"?

Occasionally there is debate that, for the sake of making daily life easy, FCL gives up some performance. This sounds logical, as in absence of a manual pre-bolus, bg values tend to go up steeper **right after meal start**. So it could be a hypothesis that, if using HCL, even better %TIR performance would have been reached.

However, the few available studies all point in the direction that FCL and HCL give similar results (see e.g. the study quoted at.

<https://androidaps.readthedocs.io/en/latest/AdvancedOptions/FullClosedLoop.html#what-to-expect>).

It seems that a well tuned-in FCL manages **late meal phases** with better results than a Hybrid Closed Loop that must pick up, as the user-given bolus fades out.

Limited value of studies in our field

Method comparisons are extremely difficult because they often suffer from problems like:

- * Methods differ in more than the investigated topic (e.g. FCL uses autoISF, HCL uses dynamicISF).

- 129 * Length of experience, and tuning quality, differs between the methods
- 130 * The true effect (how exactly the studied feature, and key tuning parameters, make a
- 131 difference in the results seen) is not analyzed, only rather global result data are generated
- 132 * The effects from technical failures (BT losses, false CGM values, occlusions/tunneling) are
- 133 not analyzed in the reported results
- 134 * Studied foods and lifestyle scenarios favor one method
- 135 * Too short of an observation period to cancel out random effects.

136 Certainly, the n=1 report given above does not qualify as a study. It is an observational report
137 that resulted, literally, quite by accident (losing the AAPS capable pump).

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139 If reading reports and studies facilitates the readers' ability and motivation to occasionally
140 attempt deeper analysis of their own daily data, and try one new idea, that is probably the
141 best use - especially if done interactively, inviting also feedback from others in the t1d
142 communities (and from your doctor, if she/he has the time...).